Question:

A baby is born after the 32nd gestational week by cesarean delivery. The mother suffered from gestational diabetes; however, she had no other pregnancy-related diseases and was otherwise healthy. The baby has a blood pressure of 100/58 mm Hg, heart rate of 104/min, and oxygen saturation of 88%. The child has tachypnea, subcostal and intercostal retractions, nasal flaring, and cyanosis. The cyanosis is responding well to initial administration of oxygen. The nasogastric tube was positioned without problems. Which of the following is the most likely diagnosis?

Answer Choices:

A. Neonatal respiratory distress syndrome (NRDS)

B. Tracheoesophageal fistula

C. Pneumonia

D. Congenital heart anomaly with right-to-left shunt

Reasoning + Answer:

<reasoning process here> The baby was born at 32 weeks gestation, which is preterm but not extremely preterm. The presence of tachypnea, subcostal and intercostal retractions, nasal flaring, and cyanosis suggests respiratory distress. Oxygen saturation is 88%, indicating hypoxemia, which is typical of neonatal respiratory distress.

The mother had gestational diabetes, which is a known risk factor for neonatal respiratory distress syndrome (NRDS) due to increased surfactant consumption and low surfactant production in the fetus. NRDS is most common in preterm infants due to immaturity of the lung and deficiency in surfactant.

In this scenario, the clinical presentation, especially the signs of respiratory distress and decreased oxygenation, combined with prematurity, aligns most closely with neonatal respiratory distress syndrome. A nasogastric tube placed without difficulty suggests the gastrointestinal tract is intact and there's no evidence for an esophageal anomaly or tracheoesophageal fistula. The absence of signs like clubbing, persistent high blood pressure, or a murmur makes a congenital heart defect with right-to-left shunt less likely. Pneumonia would present with a history of infection, increased respiratory effort, and an elevated white blood count, which is not indicated here.

The infantâĢĻs oxygenation is improving with oxygen therapy, which is consistent with NRDS, a condition that responds to surfactant replacement. While NRDS is strongly linked with prematurity, it can occur in some babies born at or near term, but here the preterm status and signs of distress point to NRDS as the most likely diagnosis.

Therefore, the most likely diagnosis is neonatal respiratory distress syndrome.

</reasoning process here>

<answer>A</answer>